

Micro and Nano Fabrication by Powder Metallurgy

- Recent advances in aerospace, automotive, bio-engineering, communications, computers and medical fields call for **miniaturised components of complex shapes in metals** and plastics due to the increasing demands for precision engineered components and systems.
- Powder metallurgy facilitates both for the **fabrication of micro grained bulk structure and for manufacturing small and tiny components in large numbers.**

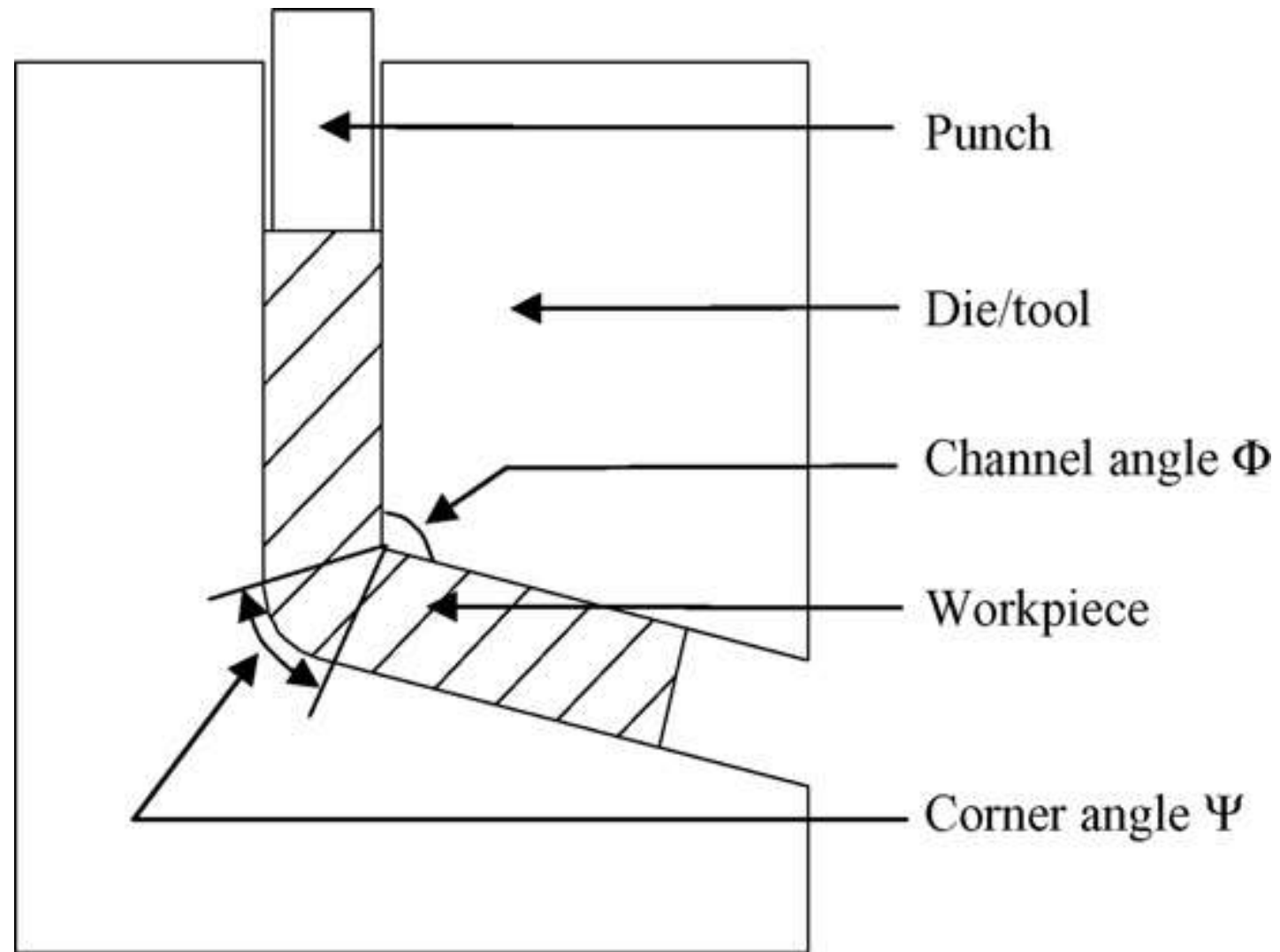
- Equal Channel Angular pressing (ECAP)
- High Pressure Torsion (HPT)
- Spark Plasma Sintering (SPS)
- Microwave sintering
- Powder Injection Moulding
- Ceramic Powder Injection Moulding (CPIM)
- Two-components CPIM (2C-CPIM)
- Micro powder injection moulding (μ PIM)
- Nano powder injection moulding (NPIM)
- Micro sacrificial plastic mould insert PIM (μ SPiPIM)
- Micro Ceramic PIM (Micro CPIM)
- Micro Two-Component Powder Injection Moulding (2C-MicroPIM).

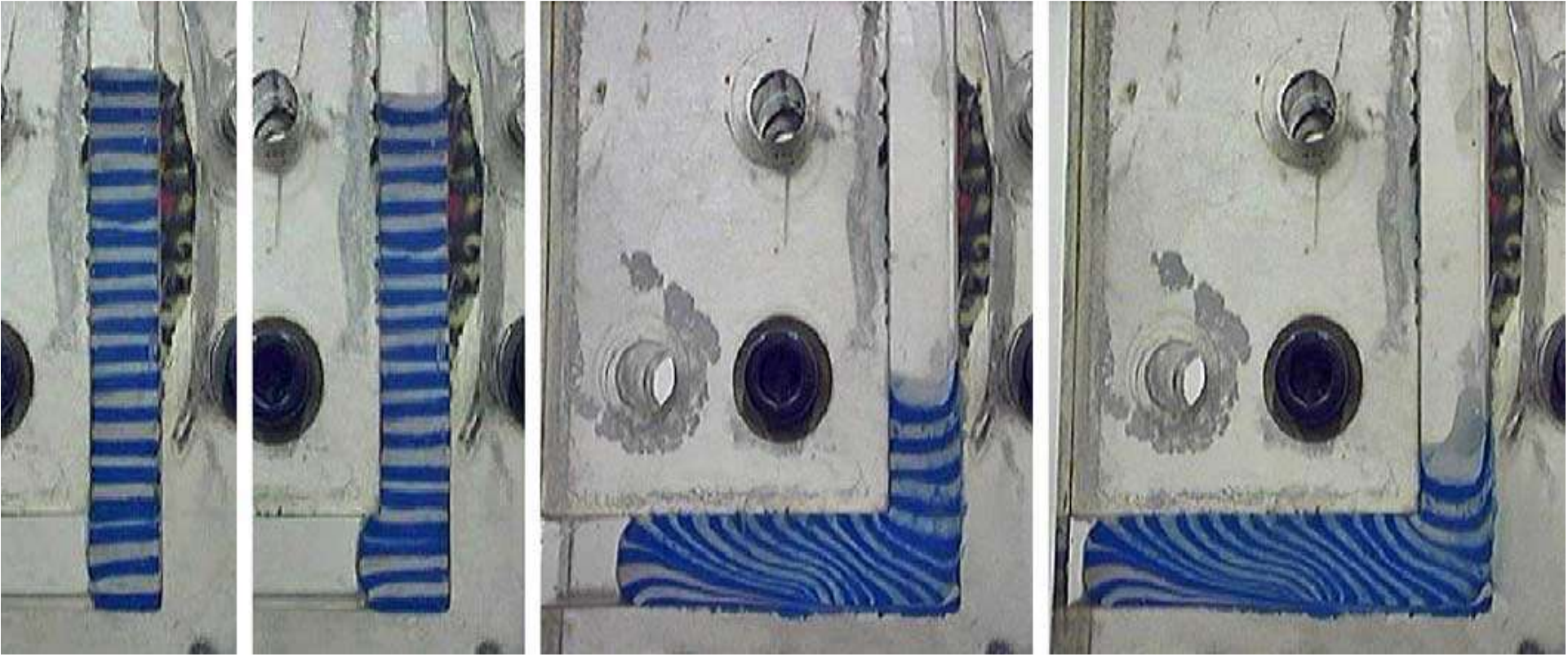
- The reason for using micro/nano powders of metals for making micro/nano structured bulk materials is due to the fact that the micro/nano particles exhibit a wide difference in characteristics in comparison with conventional powders of micron sizes.
- A bulk material of nano powders can produce completely different properties from that of micro powders.
- However, the small particles provide some advantages such as:
 - a) finer structural details, higher aspect ratios and better shape retention of microstructure
 - b) isotropic behaviour
 - c) better surface finish

- The use of **nano powders** has also an advantage of energy saving in the aspect of sintering process because of **low temperature sintering**.
- The important condition to be taken care is that, the micro/nano sized powders processed by inert gas condensation, electro deposition, ball milling, cryo milling **are not thermally stable**.
- For consolidating ultra fine grained powders for large scale structural applications, conventional pressing and sintering, Hot Isostatic Pressing, Direct extrusion employed for **consolidation of the coarse grained powders cannot be used since there is a chance for loss of ultrafine grained structure resulting in the loss of properties**.
- The crucial point in defining the **compaction or sintering temperature is the thermal stability of the powders**.
- If the temperature exceeds the thermal stability temperature either during hot compaction or sintering, the properties expected from the micro/nano grained bulk material will not be achieved.

Equal Channel Angular pressing (ECAP)

- Conventionally, Equal channel angular pressing (ECAP) is a mechanical or thermo mechanical method for the fabrication of bulk ultra fine grained or nano materials from the bulk coarse grained material by refining the grains into micro/nano size.
- The same principle is exploited for fabrication of bulk solids, in fully-dense state, from micro/nano powders.
- During ECAP, very high shear strain can be obtained which minimizes the redundant normal forces, thus yielding considerable reduction of pressing loads compared to conventional consolidation.
- In addition, it has been shown that the shear mode of plastic deformation leads to a change in the spherical pore geometry of powders into an elliptical shape, aligned with the shear plane, which is favourable for closure of voids under hydrostatic pressure.
- Since the ECAP can be carried out at room temperature or at lower temperature (below thermal stability temperature), the coarsening of the ultrafine grain size of the material can be avoided.
- Consolidation of powders by applying back pressure during ECAP has led to considerable improvement in the increase of the hydrostatic stress utilized for compaction of powders.



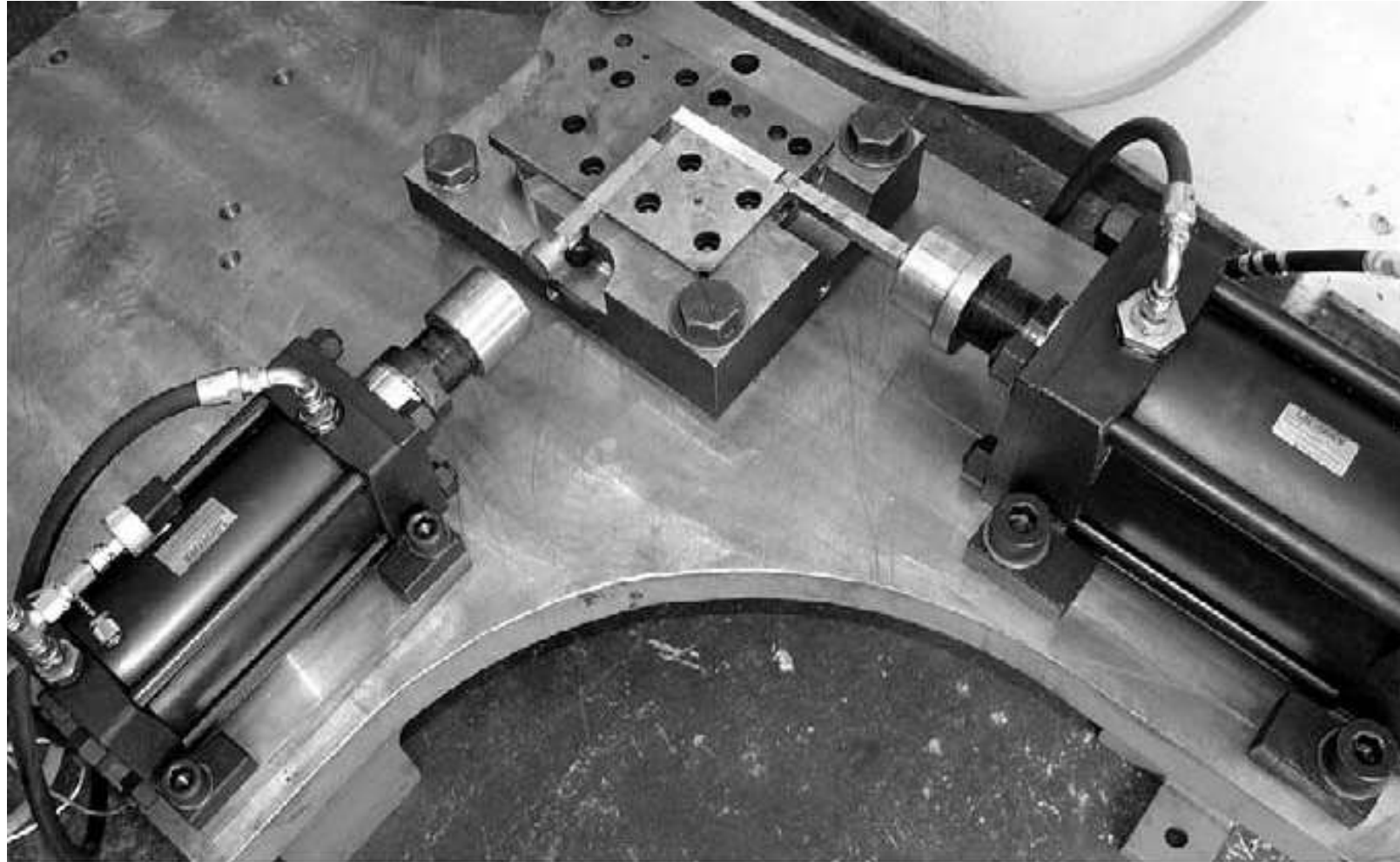


Experimental Set up

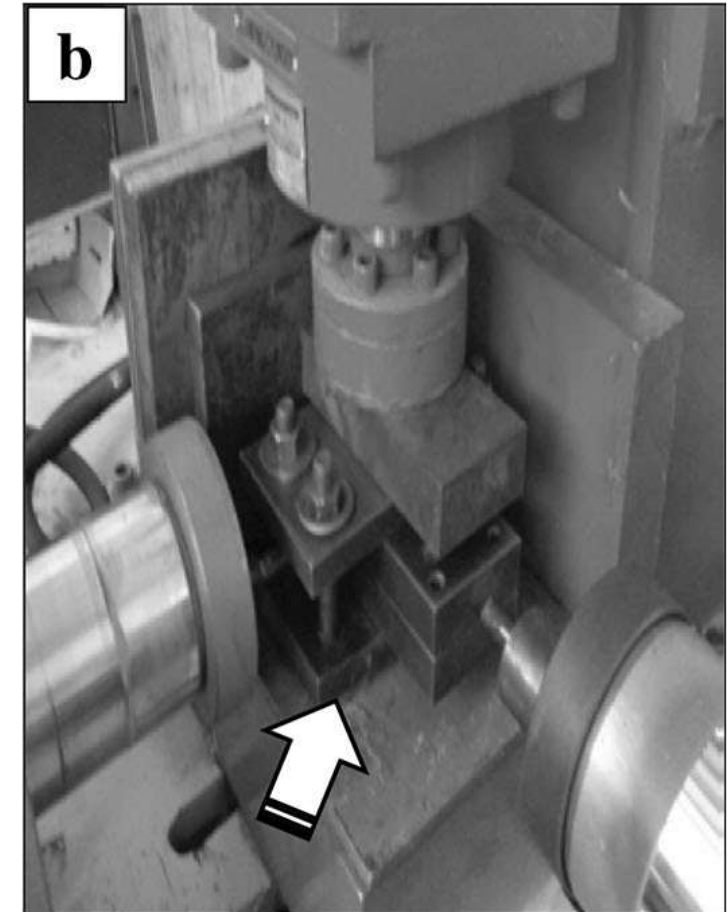
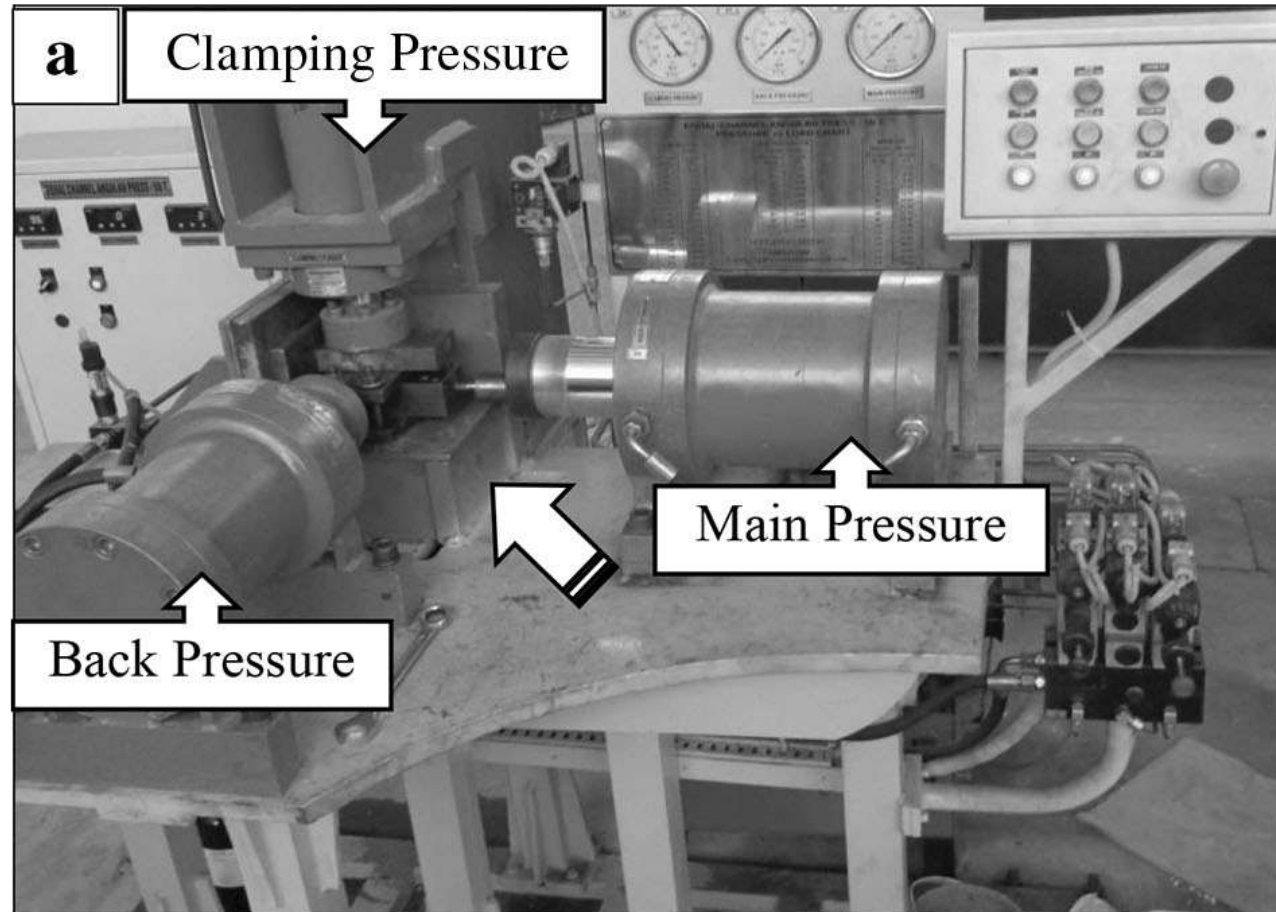


- Technically, back-pressure can be imposed in several different ways.
- The simplest way to increase back pressure within ECAP channels is to place bulk stoppers in front of the compacted materials
- The back pressure gets induced due to the obstacle in the exit side - a method widely followed.
- Better way is to apply pressure from the exit channel using a complicated die and experimental set up.

Back Pressure



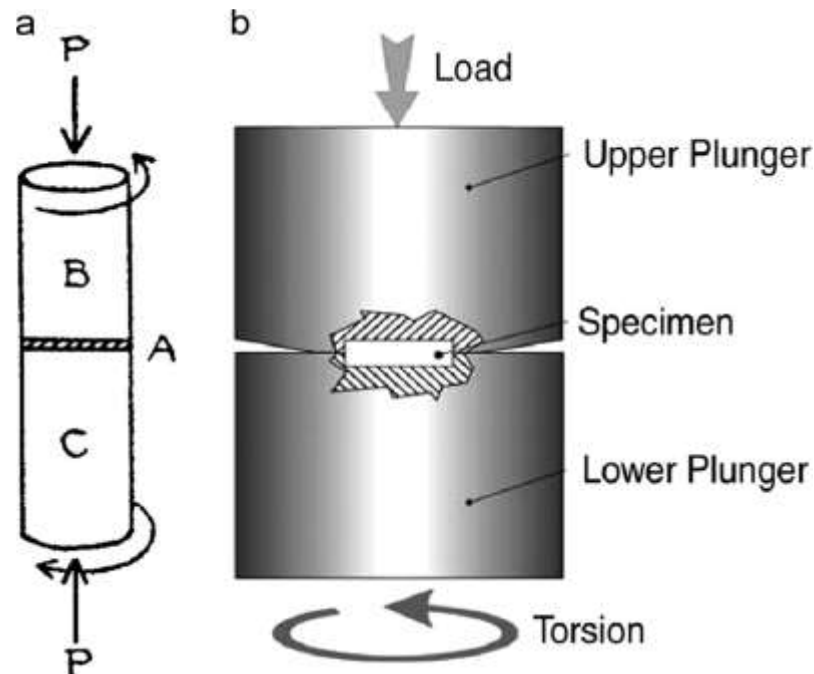
Photograph of (a) ECAP machine and (b) ECAP die set up with facility to apply back pressure.



- Several micro/nano powders have been consolidated to full density leading to bulk micro/nano structured billets through ECAP with back pressure at room temperature as well as at warm temperature.
- Micro/nano grained Al 5083, Al 2024, Copper, Titanium and their composites have been compacted to bulk nano grained materials through ECAP.

High pressure Torsion (HPT)

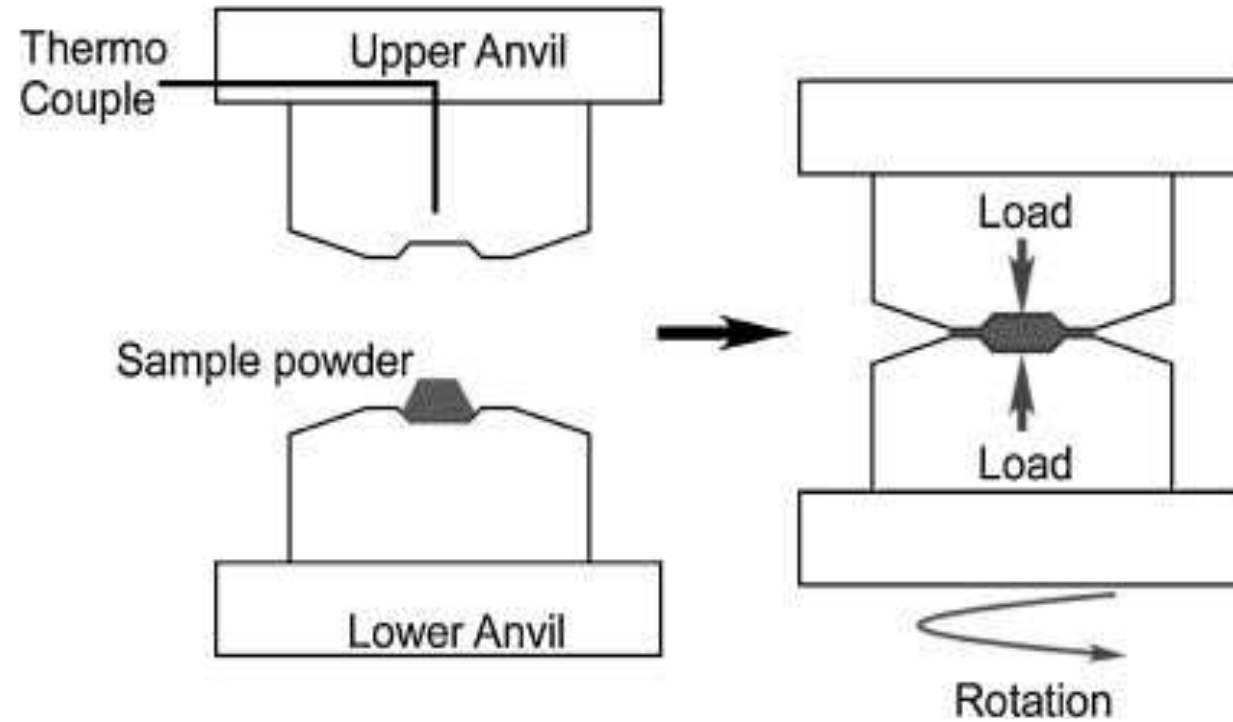
- A thin disc is squeezed between two anvils under a high pressure, P , and intense shear strain is introduced by rotating the two anvils with respect to each other.



HPT for powder consolidation

- HPT has a great potential to synthesis densely packed materials from mixed powders without a sintering process.
- It is also possible to synthesis supersaturated solid solution.
- Equilibrium solid solubility of Mg in Al is about 1 mass % at room temperature, an extended solid solubility of even up to 30 mass % of Mg in Al can be achieved by HPT of elemental powders.

Schematic illustration of HPT used for powder consolidation

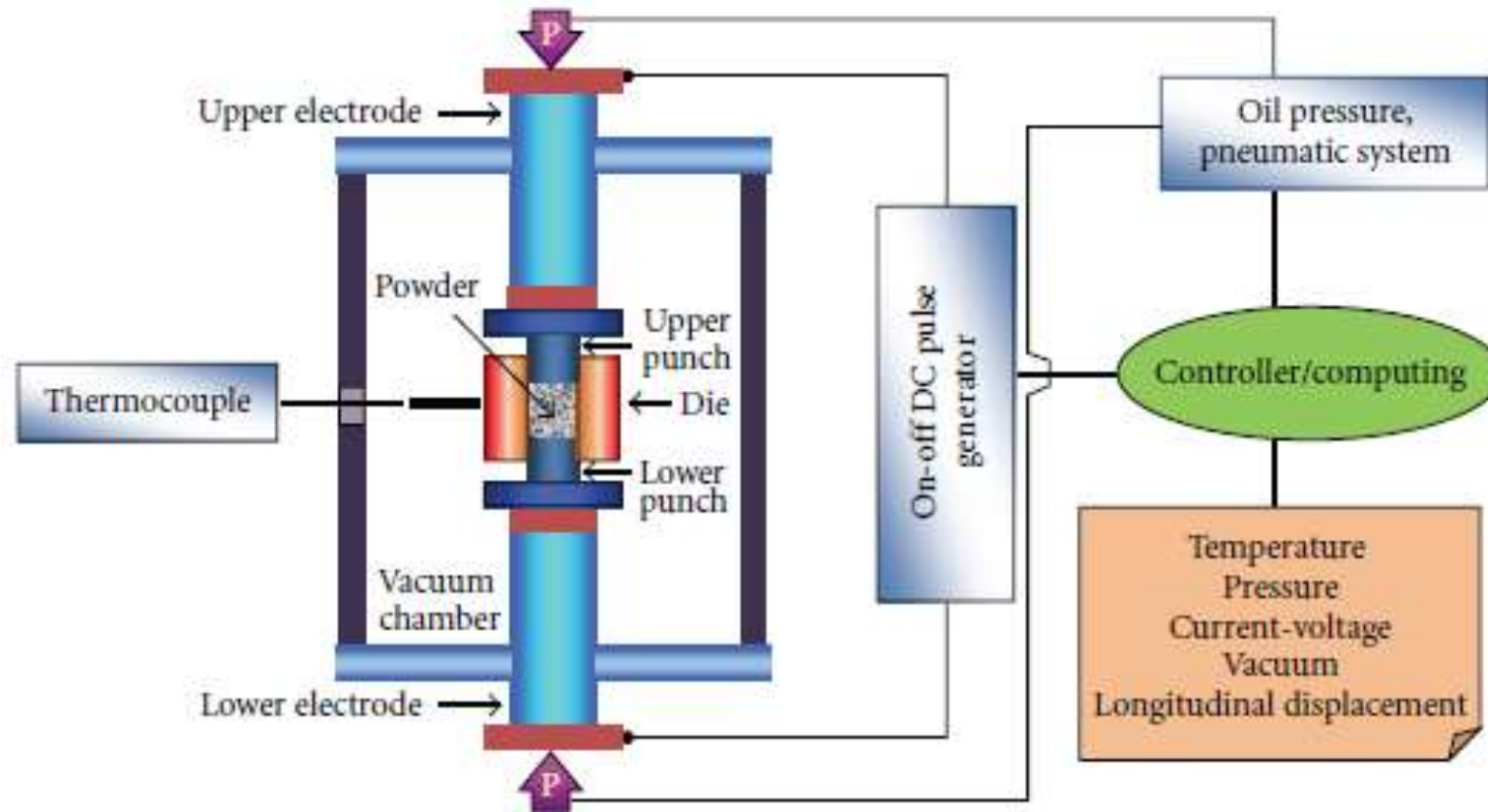


- Required amount of powder mixture is placed in a shallow circular hole, located at the center of the lower anvil of the HPT facility.
- The lower anvil is lifted until it contacted the upper anvil, which has the same dimensions as the lower anvil, and then rotated with respect to the upper anvil.
- The final sample after the HPT operation is in the shape of a disc with dimensions of circular hole.

Spark plasma sintering (SPS)

- Spark plasma sintering (SPS), also known as **field assisted sintering technique (FAST)** or **pulsed electric current sintering (PECS)**, is a technique which takes only a few minutes to complete a sintering process compared to conventional sintering which may take hours or even days for the same.
- This high sintering rate is possible in SPS since high heating rates can be easily attained **due to internal heating of the sample** as opposed to external heating used in the case of conventional sintering.
- SPS heating rates exceeding 300°C/min are easily obtained resulting in a temperature of 1200°C in only 4 minutes.
- Simultaneous application of temperature and pressure leads to high densification giving a dense compact at sintering temperatures lower by 200 to 250°C than in conventional sintering.
- In SPS, since **no coarsening and grain growth** are allowed to occur, high relative densities were reached in a very short time and nano sized powders can be sintered without considerable grain growth.

SPS Unit



- The sintering machine is assisted by a **uniaxial press, punch electrodes, vacuum chamber, controlled atmosphere**, DC pulse generator and position, temperature, and pressure measuring units.
- The **DC pulse discharge could generate spark plasma, spark impact pressure, Joule heating, and an electrical field diffusion effect.**
- In SPS, the **powder is directly fed into the graphite dies** and the die is enclosed with suitable punches unlike in conventional sintering which requires a green compact.
- This entire assembly is directly put into the SPS chamber and spacers are used if necessary.
- The chamber is now closed and the atmosphere (Vacuum, Argon etc.) in which sintering is to be carried out is applied in the chamber.
- The program is set in the control unit and sintering is carried out.
- All types of materials, even those difficult to densify can be easily sintered in SPS.
- Due to the advantage of high heating rate and less holding time, SPS can restrict the unwanted sintering reactions in highly reactive systems as opposed to conventional sintering and hence formation of undesirable product phases can be avoided.

- Three mechanisms may contribute to SPS namely, **activation of powder particles by pulsed current, resistance sintering and pressure application.**
- The high power (up to 6 kA) pulsed direct current leads to cleaning and surface activation of powders and generation of electric discharge at the neck region.
- The pulsed discharge is achieved by the application of an on/off low voltage ($\sim 30\text{V}$) and high current ($>600\text{A}$).
- The duration of each pulse varies between 1 and 300ms.
- The next step is the application of a DC current at a level dependent on the powder type.
- The pulsed and direct current may be applied simultaneously or sequentially.

- The contact between particles and the gap around the contact act as a small capacitor.
- Electrical discharges are generated across these capacitor gaps.
- Additionally, the electrical discharges around the contacts may generate plasma, that is, an ionized gas between the powder particles.
- The above phenomena collectively contribute to the physical activation of the powder particle surface.
- When spark discharge appears in the gap between the particles of a material, a local high temperature state occurs due to joule heating.
- This causes vapourisation and the melting of the surfaces of the powder particles during the SPS process resulting in the formation constricted shapes or “necks” around the contact area between the particles.
- These necks gradually develop and plastic transformation progresses during sintering, resulting in a sintered compact of over 99% density and the densification is completed within ~5-20 minutes including temperature rise and holding time.

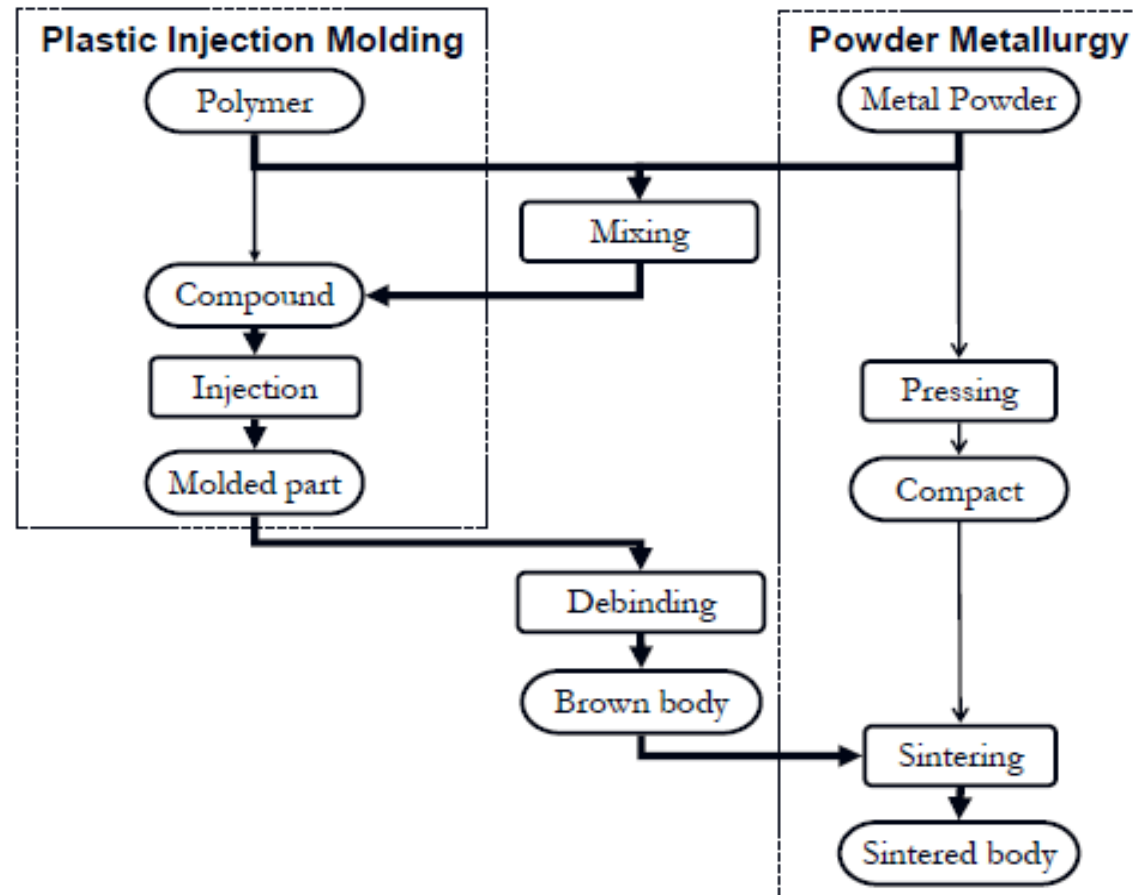
Microwave sintering

- Microwave heating is a **volumetric heating involving conversion of electromagnetic energy into thermal energy**, and is instantaneous, rapid and highly efficient.
- Most industrial activities involve microwaves at **2.45 GHz and 915 MHz frequencies**.
- Micro wave sintering is **suitable for ceramics**.
- As **metals are opaque to microwave**, metals cannot be used for heating with microwaves.
- However **metal in the form of powder absorbs microwaves at room temperature and can be heated**, with heating rates as high as $100^{\circ}\text{C min}^{-1}$, very effectively and rapidly.
- In powdered form, virtually all metals, alloys, and intermetallics will couple and heat efficiently and effectively in a microwave field, and their green parts will produce highly sintered bodies.
- The densities of microwave processed samples were close to the theoretical densities, and the net shape of the green body was preserved without significant dimensional changes.
- Thus, this technology is used to sinter various powder metal components, and has produced useful products ranging from small cylinders, rods, gears and automotive components in few minutes.

Powder Injection Moulding

- Powder injection moulding (PIM) includes metal powder injection moulding (MIM) and ceramic powder injection moulding (CIM).
- The different steps involved in Powder Injection Moulding (PIM) are **mixing of powders of the required metal, alloy, ceramic or composite with a binder material and lubricant in the form of pellets and injection moulded in a die to form the 'green' part.**
- The green part is heated in an oven for **debinding.**
- Removal of the binder is done using water or oxalic acid depending on the binder material employed.
- After debinding the **part is sintered at a suitable temperature** to bond the powder particles by diffusion so as to produce a dense near-net shaped component.

Plastic injection Moulding and powder metallurgy



Main advantages of PIM

products with higher complexity of geometrical shapes possible

high production rate

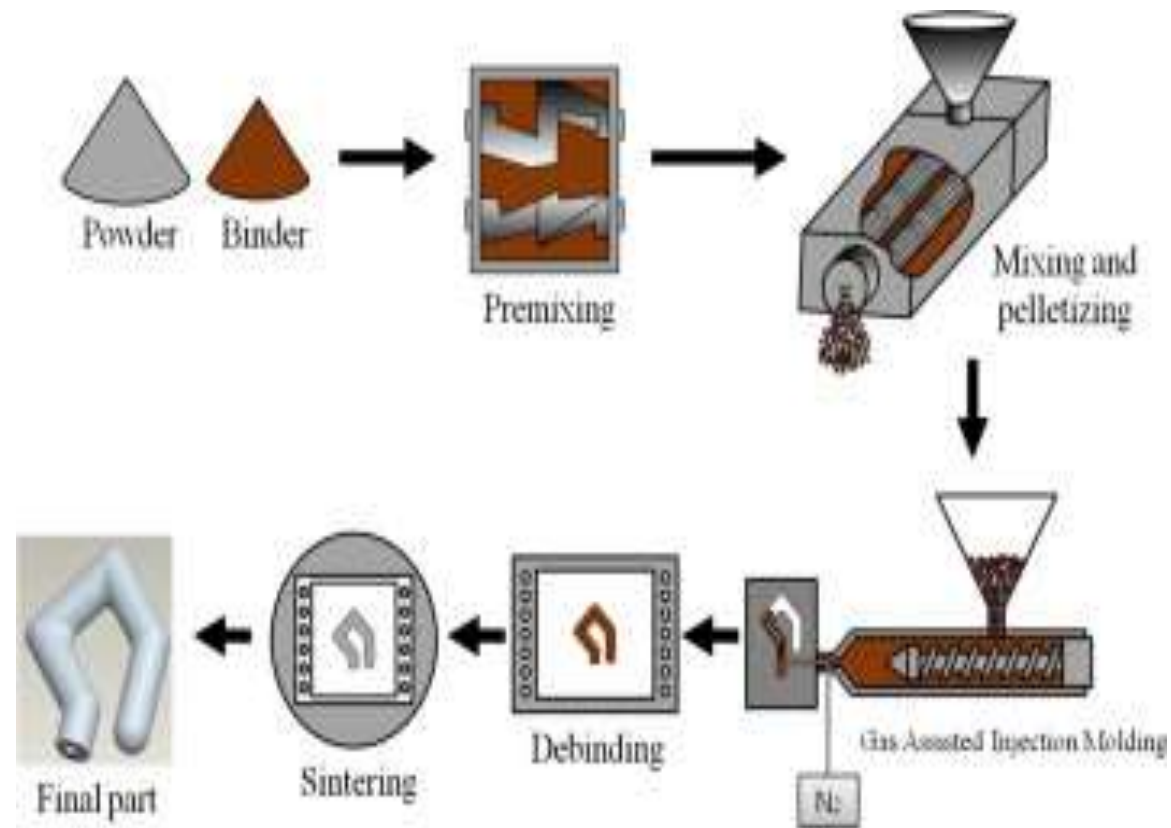
high and uniform density

repeatable higher tolerances, applicable to wide range of materials

more importantly minimal scrap loss.

three-dimensional design flexibility

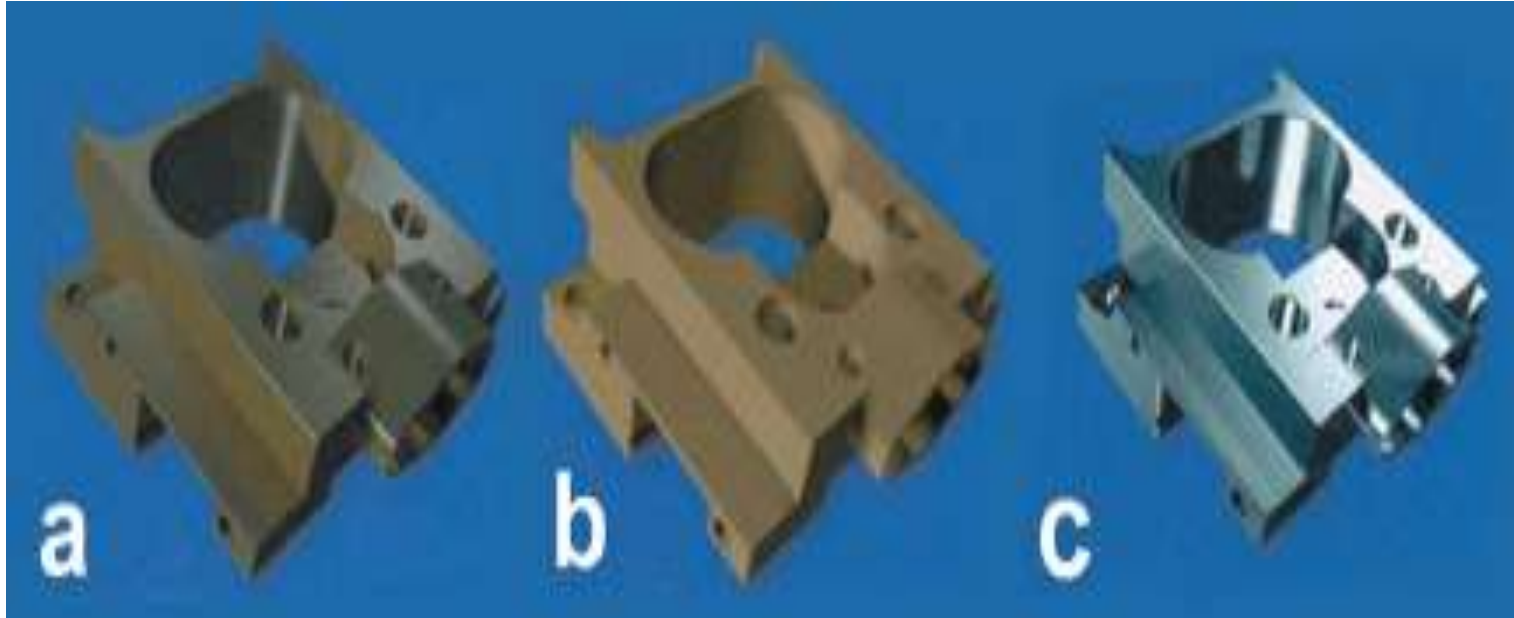
Powder injection moulding process



- Preparing the feedstock is the first step in the PIM.
- The primary raw materials for Powder Injection Molding are powders and a thermoplastic binder.
- The binder is only an intermediate processing aid to be removed from the products after injection moulding.
- The properties of the powder determine the final properties of the Powder Injection Molded product.
- The blended powder mix is worked into the plastified binder at an elevated temperature using a kneader or shear roll extruder.
- This intermediate product is the feedstock which is usually granulated with granule sizes of several millimeters, as is done in the plastic injection moulding industry.
- Feedstock can either be purchased “ready to mould” from suppliers, or it can be manufactured in-house if the necessary skills and knowledge are available with the PIM producer.

- The feed stock is formed in an injection moulding process equivalent to the production of plastic parts.
- The variety of part geometries that can be produced by this process is similar to the great variety of plastic components.
- The subsequent binder removal process produces parts with an interconnected pore network without destroying the shape of the components.
- The pore network allows the evaporation of residual binder quickly in the initial phase of sintering while the sintering necks start to grow between the metallic particles.
- The sintering process leads to the elimination of most of the pore volume formerly occupied by the binder.
- As a consequence, PIM parts exhibit a substantial shrinkage during sintering.
- The linear shrinkage is usually as high as 15 to 20%.
- If required, sintered PIM parts may be further processed by conventional metalworking processes such as heat treatments or surface treatments in the same way as cast or wrought parts.
- For certain applications, such as the automotive, medical and aerospace sectors, Hot Isostatic Pressing (HIP) can also be used to completely remove any residual porosity.

PIM part a) as moulded, b) after binder removal, c) after sintering



Metal Injection Moulding

- Metal/ alloy powders mixed with plastic binders, a plastic-bonded metal or alloy part is produced.
- Recently, **binder- less MIM** has been developed by the Boeing Company which has been granted a US Patent (20150004047A1, filed in September 2014) for binder- less Metal Injection Moulding (MIM) apparatus and method.

Ceramic Powder Injection Moulding

- The ceramic powder injection moulding (CPIM) process consists of four basic steps: feedstock preparation, injection moulding, debinding process and sintering similar to the PIM or MIM process
- The major variation is in the feed stock which contains a suitable ceramic powder and a thermo plastic binder.
- The **equipment** for injection moulding of CPIM parts are mostly similar to that used in plastic injection moulding industry except for the **use of hardened screw and barrel to reduce wear** and the modification of screw design to give homogeneity.

Low Pressure Ceramic Powder Injection Moulding

- Low pressure ceramic powder injection moulding (LPCIM) process uses pressures between 0.2 and 5 MPa and paraffin wax based binder systems.
- Low pressure injection moulding has the following **advantages**:
 - It is a simple method with low cost.
 - Due to low pressure used, it results in less wear of the machine parts leading to less contamination of the ceramic mixture by the metallic tools
 - It is easier to control the flow behaviour of the ceramic mixture
 - It also **eliminates the steps such as separate mixing, pelletising and granulation used in high pressure injection moulding**

- The LPCPIM process is capable of producing parts with complex shapes similar to casting while the mechanical properties are close to forging.
- LPCPIM is cost effective for low as well as high volumes.
- Low viscosity feedstock formulated with low-molecular weight polymers has enabled LPCPIM better mouldability and shape complexity.
- In spite of the above advantages, LPCPIM has disadvantages, the most important one is the presence of defects in the moulded parts after sintering which can be overcome by High Pressure Ceramic Powder Injection Moulding (HPCPIM) and as it can be easily automated is well suited to high volume production.

Two-components CPIM (2C-CPIM)

- Uses different combinations of ceramic materials providing the major advantage of the process namely elimination of assembling the parts after sintering.
- For the success of 2C-CPIM the chosen materials must have compatible sintering characteristics.
- The sintering control of 2C-CPIM process is crucial as both components must sinter at similar rates and similar positions in the sintering temperature to avoid delaminating.
- By lowering the powder content in one mix the sintering behaviour can be adjusted and also a non-sintering composite phase can be added.
- Depending on the design and the size of the contact area of the two ceramic materials, the feedstock of the two components can be injected either sequentially or simultaneously.

- The injection process must be performed in quick succession in order to get high compound strength.
- Also the feedstock must have comparable shrinkage rates during the co-debinding and co-sintering.
- In order to get defect-free material compounds and also to avoid high cooling stresses, the onsets of shrinkage of both components must be comparable and the thermal expansion coefficients must be almost same.
- The shrinkage of both the components must be chosen in such a way that the dynamics of densification provides a shrink fit without cracking or destruction of the assembly.
- However, by adopting a different set of parameters can result in loosening of both components and give a movable joint allowing new applications.

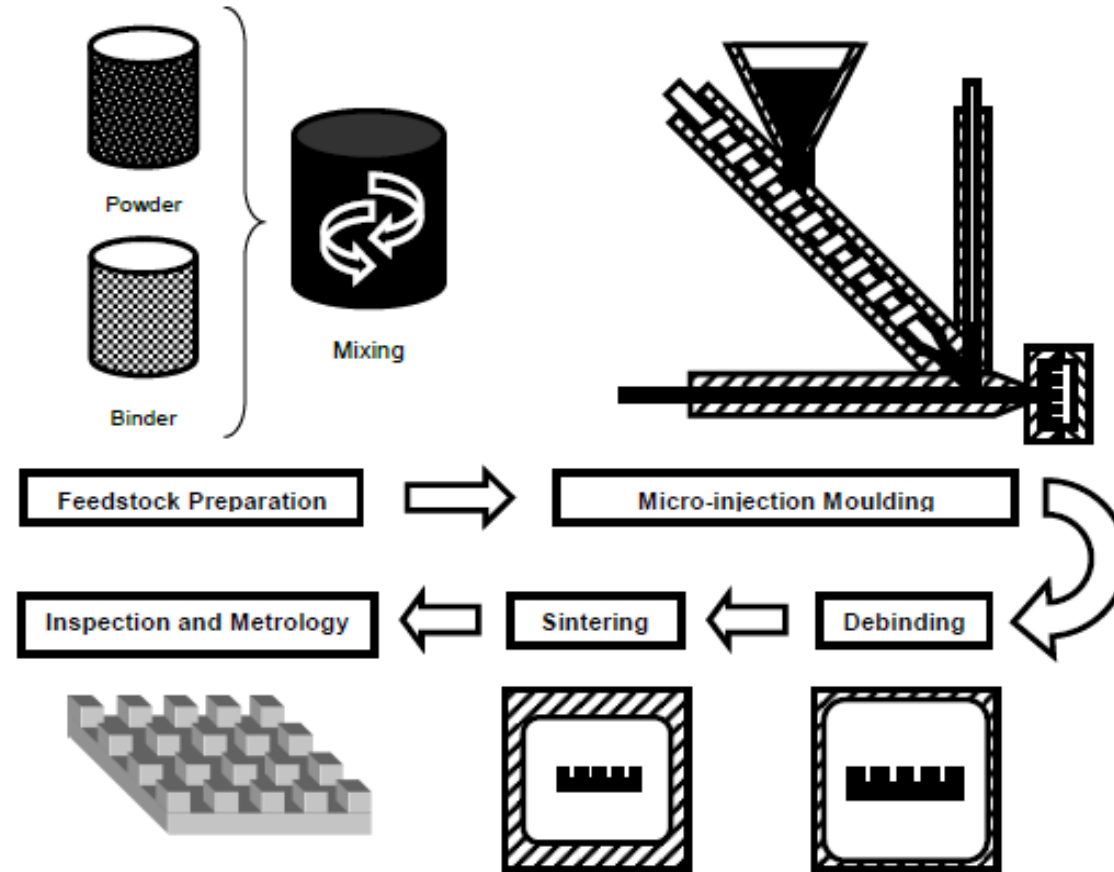
Metal-ceramic composites manufactured by 2C-CPIM



Micro Powder Injection Moulding (μ -PIM)

- Capable of manufacturing metallic and **ceramic micro-components** and systems of complex shapes with very good dimensional accuracy combined with high volume capability.
- μ -PIM is an **extension and miniaturized version of PIM** and uses **very fine particles of metal/ceramic powders** of size ranging from 2 to 5 microns.
- Unlike in the conventional PIM, μ -PIM requires **mixing of metal powders with a considerable amount of polymeric binder** which is subsequently removed after injection moulding by de-binding.
- Micro-powder injection moulding (μ -PIM) processing includes four stages: mixing of metallic powders and thermoplastic binders to prepare the feedstock, injection of powder/binder mixtures in the mould micro-cavity, thermal/catalytic or solvent debinding to remove the binder and finally solid state sintering.

Process chart for Micro- PIM



Human ear bone weighing 5.4 mg using titanium powders



Nano Powder Injection Moulding (NPIM)

- 2D and 3D structures with a height-to-width ratio of up to 10 have been recently fabricated using nano powders and a nano powder moulding process.
- Gold nano particles of 100-700 nm and platinum nano particles of 300-1000 nm.
- This nano powder moulding process is capable of fabricating nano parts for various applications.

Micro Ceramic PIM (Micro CPIM)

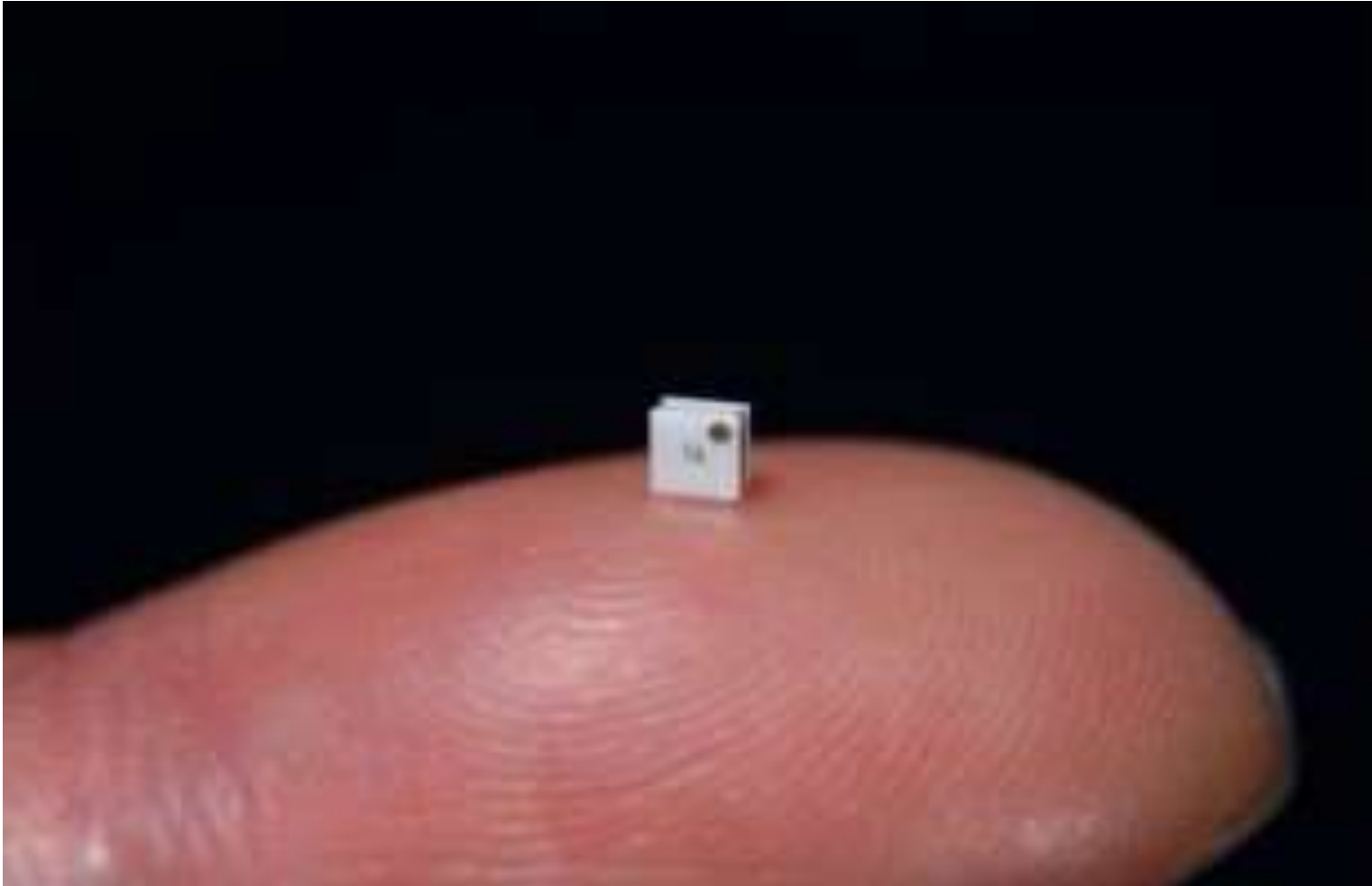
- Combination of plastic micro-injection moulding technology and ceramic powder injection moulding technology.
- While metallic materials in sub-micron sizes are mostly pyrophoric and difficult to handle, sub-micron size ceramic powders are rather easier to handle.
- Ceramic materials like ZrO_2 , Al_2O_3 , Si_3N_4 , AlN , and PZT can be formed by Micro CPIM which has applications in the field of microfluidics, biosensors, MEMS, micro system technology and medical technology.
- The growth of micro system technology gave rise to a great demand for the production of high-quality low-cost 3D micro-sized components such as micro-sensors, micro-reactors as well as other micro-parts.
- Currently available microsystem production technologies such as micro-cutting, laser ablation, LIGA are of low efficiency, high cost and are applicable to limited materials while Micro-CPIM, a variant of CPIM- technology offers a vast shape complexities and is applicable to a wide range of materials.

- Micromechanical components produced by Micro-CPIM can replace plastic parts due to corrosion resistance and high temperature performance of the ceramic materials.
- Micro parts of ceramics are used in microfluidics, micro reaction technology, and biomedical industry.
- Micro-CPIM components are used in applications such as reactions of highly corrosive media or high temperature gas phase reactions due to their hardness and high chemical and thermal resistance and may also find application for biocompatible materials.

- Micro CPIM, named after the size of injection moulded parts, can be **done either by high pressure or low pressure injection moulding** depending on the binder system.
- Micro CPIM parts weighing as low as 0.4 milligrams find applications involving high temperatures and caustic atmospheres in place of metallic or thermoplastics.
- Micro CPIM is capable of producing **parts having 20 micron wall thickness and aspect ratios up to 20 with structural details in the range of 20 to 50 microns** having surface roughness of less than 0.05 microns.

- The powder has to be **homogeneous** and in order to obtain isotropic behaviour and the **grain size of the sintered part should be at least one order of magnitude smaller than the minimum internal dimensions of the micro-part.**
- Ceramic powders with mean particle diameter of 0.5 micron or less give the best surface quality.
- The viscosity of the melt should be sufficiently low to fill even the smallest structural details in submicron range.
- When **electrostatic forces exceed gravitational forces the micro components stick to handling systems instead of dropping.**

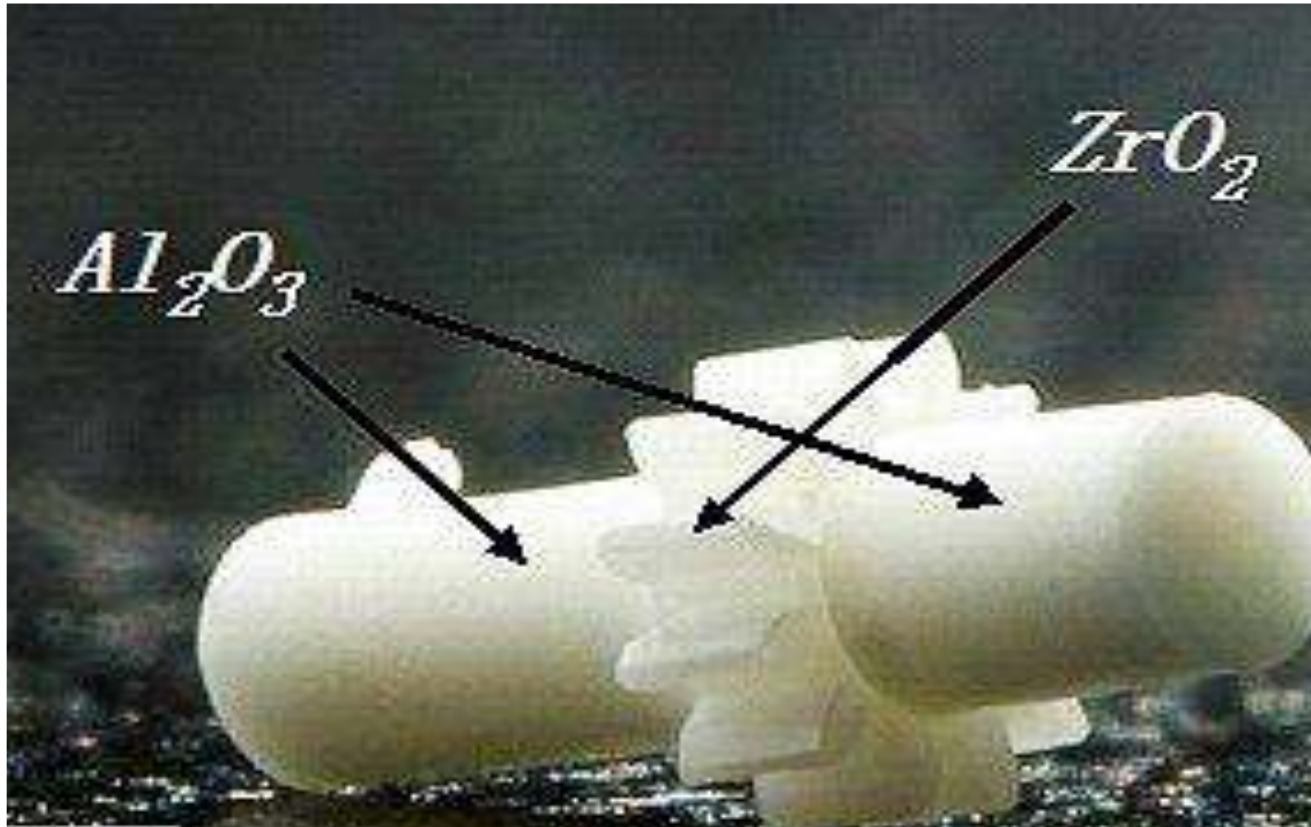
Micro switch made by Micro CPIM process



Micro Two-Component PIM (2C-MicroPIM)

- Manufacturing method for integrating two different plastics and the natural extension was **two-component injection moulding (2C-IM)** in more **resistant materials like ceramics or metals (2C-PIM)**.
- It allows for **component assemblies** to be moulded as separate low-complexity parts which are then joined into complex assemblies.
- This procedure considerably **reduces the time and cost** required to manufacture the injection moulding tools.
- Micro-Two Component Powder Injection Moulding (2C-MicroPIM) is used for the production of **highly integrated devices**.
- A **well-defined combination of powder and feedstock** is a prerequisite for **successful process set-up**.

Micro gear wheel (ZrO_2)/ shaft (Al_2O_3)



Micro Sacrificial Plastic Mould insert MIM (μ -SPiMIM)

- The major problem in micro MIM is the filling of fine details in the metallic moulds.
- However, filling of feedstock could be accomplished with lower injection pressure by using plastic mould.
- This is because the plastic mould cools the feedstock much slower than metallic one.
- Therefore, it can be expected to easily fill the viscous feedstock in narrow cavity of micro parts.
- In particular, with injection moulding of micro pillar-structured parts, it is effective to use plastic mould.
- A variety of methods for fabricating the sacrificial plastic mould such as rapid proto-typing, plastic injection-moulding, LIGA process and nanoimprint lithography process are now available.

Processing steps

- Manufacturing of SP-mould
- Injection moulding of MIM feedstock into SP-mould insert, demoulding the green compact and SP-mould insert part as one component
- Debinding to eliminate the SP-mould and polymeric binder followed by sintering process.

Nano Imprint Lithography (NIL) / μ -SPiMIM process

